

DS	Session	Arrays	Question Information	Level 1 Challenge 21
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Problem Description:

How many Y's did a Roman Centurion make a day in cold hard Lira? About a C's worth! Turns out, Martians gave Rome the idea for their number system. Use the conversion charts below to help translate some Martian numbers!

Note, that unlike the Roman Numerals, Martian Numerals reuse symbols to mean different values. B can either mean '1' or '100' depending on where it appears in the number sequence.

Input Format:

You will receive a list of numbers in a data file, one number per line, up to 5 lines at a time (with a minimum of 1 line). No number will exceed 1000, or be less than 1.

Output Format:

Print the output in a separate lines contains convert the numbers from Arabic (1,2,3...10...500...1000) to Martian (B,BB,BBB...Z...G...R) numerals.

▼ Logical Test Cases

Test Case 1	Test Case 2
INPUT (STDIN)	INPUT (STDIN)
26	161
67	627

186
408
500

EXPECTED OUTPUT

ZZWB
PZWBB
BPZZZWB
BGWBBB
G

816
401
501

EXPECTED OUTPUT

BPZB
GBZZWBB
GBBBZWB
BGB
GB

✓ Mandatory Test Cases

Test Case 1

KEYWORD

char buf[]

Test Case 2

KEYWORD

buf[i++]='R';

Test Case 3

KEYWORD

while(n>=10)

✓ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

18

Test Case 2

TOKEN COUNT

455

Test Case 3

NLOC

84

```
1 #include <stdio.h>
2 #include <string.h>
3
4 void martian_conv(int num) {
5     char buf[100];
6     int i = 0;
7     int values[] = {1000, 900, 500, 400, 100, 90, 50, 40, 10, 9, 5, 4, 1};
8     char *symbols[] = {"R", "BR", "G", "BG", "B", "ZB", "P", "ZP", "Z", "BZ", "W", "BW", "B"};
9
10    for (int j = 0; j < 13; j++) {
11        while (num >= values[j]) {
12            int len = strlen(symbols[j]);
13            for (int k = 0; k < len; k++) {
14                buf[i++] = symbols[j][k];
15            }
16            num -= values[j];
17        }
18    }
19    buf[i] = '\0';
20    printf("%s\n", buf);
21}
22
23int main() {
24    int n;
25    int count = 0;
```

```
26  while (scanf("%d", &n) != EOF && count < 5) {  
27      martian_conv(n);  
28      count++;  
29  }
```

```
30     return 0;  
31}  
32
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

26

67

186

408

500

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array1.exe

ZZWB

PZWBB

BPZZWB

BGWBBB

G

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

161

627

816

401

501

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Array1.exe

BPZB

GBZZWBB

GBBBZWB

BGB

GB

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>